

Project Plan: Visual Document Understanding (VDU) using Large Multimodal Models (LMM)

1. Project Overview

Goal: Build a proof-of-concept (PoC) system that can visually understand business documents (forms, invoices, receipts) using a Large Multimodal Model (LMM), demonstrating the potential to automate document analysis without traditional OCR.

2. Use Case

Initial focus:

- Key-value extraction from forms (e.g., "Invoice Number", "Total Amount", "Supplier Name")
- Document Q&A (e.g., user uploads a document and asks "What is the due date?")

3. Tools and Frameworks

Language: Python

Deep Learning: PyTorch, HuggingFace Transformers

Model Serving: Streamlit, FastAPI (for web/API demo)

Visualization: Streamlit, Gradio

Cloud Deployment: AWS SageMaker / GCP Vertex AI / Azure ML

Data Storage: AWS S3 / Azure Blob / GCP Cloud Storage

4. Model Options

- Donut (NAVER AI): OCR-free document understanding.
- BLIP-2 (Salesforce): Vision-to-language transformer.
- LayoutLMv3 (Microsoft): Best for layout-aware tasks.
- Pix2Struct (Google): Vision-to-text tasks.

Selected Model for Phase 1: Donut + BLIP-2 (optional phase 2)

5. Dataset Choices

- FUNSD: Form key-value extraction
- SROIE: Receipts
- CORD: Korean receipts
- DocVQA: Document Q&A

Selected Dataset: FUNSD

6. Phase-by-Phase Plan

Phase 1: Setup and Experimentation (1 week)

- Install Donut, run on sample images
- Visualize with Streamlit

Phase 2: Fine-tuning (1-2 weeks)

- Load FUNSD, fine-tune Donut
- Evaluate with accuracy, F1-score

Phase 3: Optional Q&A Module (1 week)

- Integrate BLIP-2 or LLaVA
- Test with DocVQA

Phase 4: Cloud Deployment (1 week)

- Deploy to AWS SageMaker or GCP Vertex AI
- Build Streamlit or FastAPI front-end

Phase 5: Documentation & Demo (1 week)

- Write documentation, record demo, prepare pitch deck

7. Stretch Goals (Future Work)

- Multilingual docs, LoRA for training, classification, mobile OCR

8. Deliverables

- Trained LMM model, web demo, cloud endpoint, documentation

9. Evaluation Metrics

- JSON accuracy, BLEU/F1-score for Q&A, usability feedback

10. Next Steps

- Start Phase 1: Install Donut, run inference, setup FUNSD
- Schedule checkpoints after each phase